

Name _____ Period _____

Skill 30.1 Exercise 1

Refer to the *titanic* dataframe below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

In the space below, make crosstab table that compares *Sex* to *Survived*

Convert the frequencies in the crosstab table below to proportions.

The actual crosstab chart which compares Sex and Survived as proportions is shown below.

Survived	0	1
Sex		
female	0.090909	0.261504
male	0.525253	0.122334

Based on the proportions in the table, does survival appear to be associated with sex? Justify your answer using specific values.

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Skill 30.2 Exercise 1

In the previous exercise, we calculated the proportions for the survivors and sex.

Survived	0	1
Sex		
female	0.090909	0.261504
male	0.525253	0.122334

Calculate the marginal proportions below (row totals and column totals)

Survived	0	1	Row totals
Sex			
female	0.090909	0.261504	
male	0.525253	0.122334	
Column totals			

Calculate the expected proportions,

Survived	0	1
Sex		
female		
male		

Calculate the expected frequencies by multiplying proportion by the sample size (891 for this data)

Survived	0	1
Sex		
female		
male		

What does the table above show?

The actual frequencies are shown below,

Survived	0	1
Sex		
female	81	233
male	468	109

Compare the calculated frequencies to the actual frequencies. What do these tables tell us about the association between these two variables?

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Skill 30.2 Exercise 2

Use the *chi2_contingency* function to calculate the expected counts and print the results

Skill 30.3 Exercise 1

Refer to the *chi2_frequency* function you applied in the previous example. Print the *chi2* statistic.

Fill in the blank below.

For a 2x2 table (like the one we've been investigating), a Chi-Square statistic larger than around _____ would strongly suggest an association between the variables.

The actual *chi2* statistic is 261. What does this value tell us about the association of the *sex* and *survivor* variables?